Chapter 3 lecture script: From samples to estimates

Draft for the course author's review. **Target: 150 minutes in total, with 125 minutes of slides on 34 slides (including the questions and discussions below), 5 minutes of slack and two breaks of 10 minutes.** The breaks separate the three blocks of the lecture. There are no live demos: whether to run code is decided during the lecture according to the time left, and "Adjustments during delivery" lists the chunks that fit.

This script follows Chapter 3 theory, in its order, with a short introduction to the R practical. The slides carry keywords only; the narrative is here.

Slides 3 and 4 use your diagrams `ModelCycle.png` and `Estimation.png` from the Chapter 1 lecture folder. Slide 17 uses the schematic `hypothetical-samples.svg`. Slides 8 (`relative-frequency-convergence.png`), 9 (`fir-empirical-distribution.png`), 12 (`histogram-density.png`), 13 (`fig-ecdf_normal-1.png`), 19 (`sampling-distribution.png`), 26 (`dance-ci.png`) and 30 (`asymptotic-normality.png`) use copies of the chapter figures `fig-relative-frequency-convergence`, `fig-ecdf`, `fig-histogram_density`, `fig-ecdf_normal`, `fig-sampling_distribution`, `fig-dance_ci` and `fig-asymptotic_normality`. All are in `lecture-assets/`, and the code of the PNG figures, except the one of slide 13, is in `lecture-assets/figures.R`. If the chapter figures change, run that script again and replace the copy of slide 13 by hand.

Timing and slide sequence

Times are measured from the start of the session and include the breaks.

Slide	Topic	Minutes	Elapsed time
1	Ecological Models and Data Analysis	1	0–1
2	Overview of the lecture	2	1–3
3	Where we are: the modelling cycle	1	3–4
4	From data to estimates	2	4–6
5	Samples and distributions	4	6–10
6	From frequency to probability	5	10–15
7	Relative frequencies in R	3	15–18
8	The law of large numbers	5	18–23
9	The empirical distribution	4	23–27
10	Summaries of a sample	5	27–32
11	Samples of continuous measurements	3	32–35

Slide	Topic	Minutes	Elapsed time
12	Density-scaled histograms	6	35–41
13	The empirical cumulative distribution	3	41–44
	Break	10	44–54
14	Parameters, estimators and estimates	4	54–58
15	The method of moments	4	58–62
16	Discussion: parameter, estimator or estimate?	4	62–66
17	The sampling distribution	6	66–72
18	Hypothetical samples in R	3	72–75
19	A simulated sampling distribution	3	75–78
20	Bias and variance	6	78–84
21	Mean squared error	3	84–87
22	Standard errors	5	87–92
23	Discussion: standard deviation or standard error?	4	92–96
	Break	10	96–106
24	Confidence intervals	4	106–110
25	Wald interval in R	2	110–112
26	The dance of the confidence intervals	5	112–117
27	Coverage checked by simulation	4	117–121
28	Estimation of two parameters	5	121–126
29	Consistency	3	126–129
30	Asymptotic normality	5	129–134
31	Discussion: empirical or sampling distribution?	3	134–137
32	Course roadmap: from samples to response curves	2	137–139
33	The Chapter 3 R practical	2	139–141
34	Summary and closing questions	4	141–145
	Slack for discussion	5	145–150
	Total	**150**	

Slides 125 minutes (blocks of 44, 42 and 39), slack 5, breaks 20: 150 minutes.

Slides 1 and 2 — Ecological Models and Data Analysis; Overview of the lecture

In this lecture, we learn how to relate samples of data to probability distributions and how to use a sample to estimate the parameters of a model. The session has three blocks, separated by two breaks.

The first block is about samples: how the relative frequencies of a sample of counts relate to the probabilities of a model, and how we describe a sample of continuous measurements with histograms and cumulative distributions.

The second block turns the question around. We define parameters, estimators and estimates, calculate a first estimate with the method of moments, and ask how much that estimate would change with another sample. That leads to the sampling distribution, bias, variance, mean squared error and standard errors.

The third block covers confidence intervals and their coverage, the estimation of two parameters, and what happens to estimators as the sample grows. It ends with the roadmap, the practical and the summary.

The slides show only part of the code. The calculations are all on the Chapter 3 page, where you can run them yourselves.

Slides 3 and 4 — Where we are: the modelling cycle; From data to estimates

In Chapter 2 we chose the parameters of a distribution and calculated probabilities from them: the distribution in the model box of the cycle. This chapter is the first step of estimation: "confidence intervals (3)" in the estimation box. Chapter 4 continues with the likelihood.

On slide 4, the diagram from Chapter 1. In Chapter 2 we went from parameters to data: the parameters were known and we calculated the probabilities of possible observations. In research we face the reverse problem: we have a limited sample of observations and want to learn about the parameters of the model that we use to describe them. The dashed arrow from the estimates to the parameters is the relationship that this lecture makes precise: how close an estimate is likely to be to the parameter, and how we express that.

Delivery

One minute on slide 3 and two on slide 4. On slide 3, point to "Confidence intervals (3)" only; do not walk through the cycle again. On slide 4, point to the dashed arrow from the estimates to the parameters.

Slide 5 — Samples and distributions

In research we collect limited, often noisy samples of data and want to generalise from them to the broader context in which they were collected. As discussed in Chapter 1, we use statistical models as approximations of the data generating process. This means that we treat the observations that

we collect as if they were random samples from our statistical model. For example, we might want to learn about the density of tree seedlings in a forest while counting them only in some quadrats. To make that connection, we need assumptions about how the quadrats were selected and how their counts vary from quadrat to quadrat, and we describe that variation with a distribution that is (at least qualitatively) compatible with the process.

The example of the chapter is a set of counts of balsam fir seedlings in 50 quadrats of five by five feet, laid out in a grid of five rows of ten quadrats (Davison and Hinkley 1997, *Bootstrap Methods and Their Application*, Cambridge University Press). The data come with R, as the data set `fir` of the package `boot`. The number of seedlings in a quadrat has no fixed upper limit, so we use a Poisson distribution with a mean of 2.14 seedlings per quadrat. That value is the mean of the 50 counts; slide 15 explains why it is a reasonable value for the parameter. A Poisson distribution assigns a probability to every count, however large, but with this mean the probability of more than eight seedlings in a quadrat is below 0.001, so the comparisons stop at eight.

We treat the 50 counts as independent and identically distributed (i.i.d.). Neighbouring quadrats of a grid may be more alike than distant ones, so this is a simplification. The course does not model this spatial dependence; Chapter 7 deals with a related case, observations that come in groups.

Slide 6 — From frequency to probability

For a sample of counts, the **relative frequency** of a value is the number of times it appears divided by the sample size. The table puts the relative frequencies of the 50 quadrats beside the probabilities of the Poisson model. No quadrat has six or eight seedlings. The relative frequencies lie between zero and one and sum to one, like probabilities.

The two rows are similar, but they are not equal. For small samples, relative frequencies and model probabilities can differ greatly; the next slides show what happens as the sample grows.

Short activity — included in the five minutes

Ask the question on the slide and take two answers. Expected answers: part of the difference is sampling variation. Even if the model were an exact description of the data generating process, a sample of 50 quadrats would not reproduce its probabilities; another 50 quadrats would give other frequencies. The other part is that the Poisson model is an approximation of the process that produced the counts (and the quadrats may not be independent). From one sample we cannot tell how much of the difference is due to each.

Slide 7 — Relative frequencies in R

These are the chunks of the chapter that produce the table of slide 6. `data()` loads the data set from the package `boot`, and the counts are the column `count`. `table()` counts how often each value appears. Some possible counts may be absent from a sample, and `table()` would then leave them out; setting the values from zero to eight as `factor()` levels makes `table()` include those counts with frequency zero. Dividing by the number of quadrats gives the relative frequencies, and `dpois()` gives the model probabilities for the same counts.

As in Chapter 2, the distribution functions come from the packages RTMB and RTMBdist, which the chapter loads once, RTMB first.

Slide 8 — The law of large numbers

For i.i.d. observations, the relative frequency of each count approaches its model probability as the sample size grows. This result is known as the **law of large numbers**. It is a statement about probabilities and not about every sample: as the sample grows, a large difference between the relative frequency and the model probability becomes less and less likely, but a larger realised sample need not be closer to the model probability than every smaller realised sample.

With real data we cannot make the sample larger, but in a simulation we can. The chapter simulates one sample of each size from 10 to 100,000 quadrats with seed 123. One sequence of samples cannot show convergence directly: in that table, for example, the relative frequency of two seedlings is closer to the model probability at 100 quadrats than at 10,000. To show what usually changes with sample size, the figure repeats the simulation 100 times at each size.

Figure walkthrough — included in the five minutes

The figure is a copy of the chapter figure. Each grey point is the relative frequency of exactly three seedlings in one simulated sample; the horizontal axis is the sample size on a logarithmic scale, and the vermillion line is the probability of three seedlings under the Poisson model with a mean of 2.14. At 10 quadrats the relative frequency can only be a multiple of 0.1, and the points range from 0 to 0.5. As the sample size grows, the points concentrate around the line. Individual samples can still fall far from the line, but large deviations become less likely.

Optional remark

The law of large numbers holds for one event at a time. Glivenko and Cantelli proved in 1933 that the empirical cumulative distribution of an i.i.d. sample (next slide) converges to the cumulative distribution of the model for all values at once. The formal statements are in the chapter's box "Statistical detail: the law of large numbers and the Glivenko–Cantelli theorem".

Slide 9 — The empirical distribution

Because of this relationship between relative frequencies and probabilities, we often define the **empirical distribution** of a sample: we assign each observed count its relative frequency and treat the result as if it were a probability mass function. In principle it is just a way to tabulate a (potentially long) collection of observations, but it lets us calculate properties of a sample, such as quantiles or expectations, that we can relate to the same properties of the model.

Like any distribution, an empirical distribution has a cumulative function: the fraction of observations at or below each count. When we compare the empirical and model cumulative probability functions, we often use step plots, which show both functions side by side. These plots are useful when we want to see how well a distribution matches a sample, that is, whether a sample of observations could be treated as a random sample from a given distribution. Later chapters return to this.

Figure walkthrough — included in the four minutes

The figure is a copy of the chapter figure. Left, the probability mass: the vermillion stems are the Poisson probabilities and the blue points the relative frequencies of the 50 quadrats, the two rows of the table on slide 6. Right, the cumulative versions as steps: the solid vermillion line is the model and the dashed blue line the sample. The two cumulative functions are close everywhere, although the masses differ visibly at some counts (zero and four seedlings are more common in the sample than in the model, three less common).

Optional remark (the chapter's footnote)

Andrey Kolmogorov first formalised and defined the empirical distribution function (the cumulative version) in 1933.

Slide 10 — Summaries of a sample

Just like for any probability distribution of Chapter 2, we can calculate expectations such as means and variances from an empirical distribution, and quantiles by inverting the empirical cumulative distribution: we replace the probabilities in the formulae by the relative frequencies of the sample.

Strictly speaking, a data sample is just a collection of values. We can always calculate the arithmetic mean of those values, but in this course a mean and a variance are properties of distributions. So when we say "sample mean" or "sample variance", we mean the mean or the variance of the empirical distribution. The mean of the empirical distribution equals the arithmetic mean of the sample, 2.14 seedlings per quadrat. The variance of the empirical distribution divides by the number of observations n and gives 2.36, whereas `var()` divides by $n - 1$ and gives 2.41. What most people call the sample variance is an estimator of the variance of the model, a different concept; slide 28 explains why it uses a different divisor. The median, the 0.5 quantile, is the smallest observed count whose empirical cumulative probability reaches 0.5: 2 seedlings. R has nine ways of computing a quantile, and `quantile()` with `type = 1` returns the smallest observed count whose empirical cumulative probability reaches the requested value, which is what we want for counts.

For independent observations from the same distribution, the empirical distribution converges to the model distribution as the sample size grows. Its moments (the mean, the variance and others) converge to those of the model, provided that those moments exist in the model, and a quantile converges where the cumulative probability function of the model is not flat at that quantile. All the moments of a Poisson distribution exist, so the mean and the variance of our sample would converge to those of the model if the counts were i.i.d. Poisson. A finite sample can sometimes agree exactly, but agreement is not guaranteed at any finite size.

In practice we never have an infinite sample, but the larger the sample, the better we can judge whether a model describes the data. Larger samples are also more stable: there is less variation across hypothetical samples of the same size, which is the topic of the second block. Note that agreement between a sample and a model does not show that the observations are independent; independence remains an assumption.

Slide 11 — Samples of continuous measurements

The relationship between samples and distributions is the same for continuous measurements, but the tools to describe a sample need to be adapted. A continuous random variable takes any particular value with probability zero, so in a sample of exact measurements we would not expect any value to be repeated: the relative frequency of every observed value would be $1/n$, and a table of relative frequencies would tell us nothing.

The example is the flipper lengths of the 123 Gentoo penguins of Chapter 2 (the data set `penguins`, built into R 4.5.0 and later; one penguin without measurements is removed), with the Normal model of that chapter: a mean of 217 mm and a standard deviation of 6.5 mm. The lengths were recorded to the nearest millimetre, so some values do appear more than once, but the 123 penguins are spread over 25 different values.

A **histogram** is the continuous counterpart of the table of relative frequencies. We divide the range of the measurements into intervals, called **bins**, and count the observations in each bin. Dividing the count by the sample size gives the relative frequency of the bin, which we can compare with the probability that the model assigns to that interval. We place the limits of the bins halfway between two millimetres, so that no recorded length falls on a limit. For the bin from 214.5 to 218.5 mm, 31 of the 123 penguins give a relative frequency of about 0.252, and the model probability is about 0.241. As with counts, the relative frequency of a bin converges to the model probability of the interval as the sample grows, when the observations are i.i.d. from the model.

Slide 12 — Density-scaled histograms

To compare a histogram with a probability density function we need the same units. In a **density-scaled histogram** the height of each bar is the relative frequency divided by the width of the bin, so that the area of the bar equals the relative frequency and the areas of all bars add up to one. For the bin of slide 11, the height is $0.252/4 = 0.063$/mm, against a Normal density of about 0.061/mm in the middle of the bin. In R, `hist()` draws a histogram of counts by default and a density-scaled histogram with the argument `probability = TRUE`.

The width of the bins is our choice, and it changes how the same sample looks. Narrow bins contain few observations each, so the heights of the bars vary a lot from bin to bin. Wide bins are more stable, but hide the shape of the distribution within each bin. There is no correct bin width; the larger the sample, the narrower the bins can be.

The dashed blue curve is a **kernel density estimate**, a smooth alternative to the histogram. Instead of counting observations in bins, it places a small smooth curve (the kernel, by default a Normal density) on top of every observation and averages these curves. The standard deviation of the kernel is called the **bandwidth**, and it plays the same role as the bin width: a small bandwidth gives a wiggly estimate and a large bandwidth a very smooth one. In R, `density()` calculates it and chooses the bandwidth from the sample.

A kernel density estimate is calculated from the sample without assuming a model, whereas the vermillion curve is the density of the model that we chose. The two curves differ for two possible reasons: a sample of 123 penguins varies from one sample to the next (sampling variation), and the

Normal model is an approximation of the data generating process. From one sample we cannot tell how much of the difference is due to each.

Figure walkthrough

The figure is a copy of the chapter figure: the same 123 lengths with bins of 1, 4 and 12 mm. Point to the middle panel, where the bar from 214.5 to 218.5 mm is the one calculated above, then to the left panel (bars jump up and down) and the right panel (the shape is lost).

Short activity — included in the six minutes

Ask the two questions on the slide (the chapter's "Practice: bins and samples"). Expected answers: other penguins give a different sample, so the histogram would change, although with 123 penguins its overall shape should remain similar. The bin width only changes how the sample is displayed: the values in the sample stay the same, and so does their mean.

Slide 13 — The empirical cumulative distribution

The empirical cumulative distribution needs no adaptation for continuous data: it is still the fraction of observations at or below each value, and it requires neither bins nor a bandwidth. With continuous data it increases by $1/n$ at each observation (and by a multiple of $1/n$ where several rounded measurements coincide), whereas the cumulative probability function of the model is a smooth curve. In R, `ecdf()` returns the empirical cumulative distribution of a sample as a function that we can evaluate at any value.

For a flipper length of 225 mm, 88% of the penguins have a flipper of at most that length, while the model assigns a probability of about 0.89 to that event. The mean, variance and quantiles of a continuous sample are calculated as before (for continuous data the default `type` of `quantile()` is appropriate), and the asymptotic convergence of slide 10 also holds: for i.i.d. observations from a model, the empirical cumulative distribution converges to the cumulative probability function of that model as the sample grows.

Figure walkthrough — included in the three minutes

The figure is a copy of the chapter figure. The solid vermillion curve is the Normal model and the dashed blue steps are the 123 penguins. Point to the steps, which are larger where several rounded lengths coincide, and then to 225 mm on the horizontal axis, where the two lines are close (0.88 and 0.89).

Break (10 minutes).

Slide 14 — Parameters, estimators and estimates

A **parameter** is a quantity that defines a model distribution, such as the probability p of a Binomial distribution or the mean μ and standard deviation σ of a Normal distribution. In Chapter 2 the values of the parameters were given. With real data they are unknown, and we want to learn about them from a sample.

We return to the fir seedlings. Let X_i be the number of seedlings in quadrat i. We model the counts with a Poisson distribution, where λ is the expected number of seedlings per quadrat, and we assume that the counts are i.i.d. That is, all quadrats have the same area and the same expected count, and the count in one quadrat does not change the probabilities for the others. The parameter λ is unknown.

An **estimator** is a rule (a function) that we apply to a sample to calculate a value for a parameter. Since we treat the observations as random variables, an estimator is also a random variable: a different sample would give a different result. An **estimate** is the value that the estimator returns for the sample that we actually observed. We mark estimators and estimates with a hat, for example $\hat{\lambda}$ for the parameter λ.

Slide 15 — The method of moments

The mean of the 50 counts is 2.14 seedlings per quadrat. As λ is the mean of the Poisson distribution, the mean of the sample seems a reasonable estimate of λ, and it is the value that we gave to the model on slide 5. The **method of moments** turns this intuition into a general procedure: it equates expectations of the model distribution to the same expectations of the empirical distribution of the sample, and solves the resulting equations for the parameters. A model with one parameter needs one equation, normally the mean; a model with two parameters needs a second equation, normally the variance (slide 28).

For a Poisson distribution $E[X_i] = \lambda$, and the mean of the empirical distribution is the mean of the n counts, $\overline{X}$. Equating the two gives the estimator $\hat{\lambda} = \overline{X}$, and for the 50 quadrats the estimate is 2.14 seedlings per quadrat.

The justification for the method is the convergence of slide 10: if the moments of the empirical distribution converge to the moments of the model as the sample grows, a parameter calculated from the former should be close to the parameter of the model when the sample is large enough.

Maximum likelihood (Chapter 4) is the method of estimation that we use in the rest of the course. We start with the method of moments because its estimators are short formulae, which makes it easier to study how estimates vary across samples. The concepts of the rest of this lecture (sampling distribution, standard error and confidence interval) apply to any estimator.

Optional remark (the chapter's footnote)

Formally, the method of moments equates the raw moments $E[X^k]$ of the model and the empirical distribution. The central moments are functions of the raw moments, so equating them gives the same estimators; the chapter uses central moments because the mean and the variance are the summaries that we are used to interpreting.

Slide 16 — Discussion: parameter, estimator or estimate?

This discussion checks the three terms of slide 14 before we use them in the rest of the block.

Discussion and expected answers — included in the four minutes

Give students one minute, then take one answer per item.

1. **Parameter:** λ is a fixed, unknown quantity of the model.
2. **Estimator:** a rule that we apply to a sample; here it is $\hat{\lambda} = \overline{X}$.
3. **Estimate:** the number that the rule gives for the sample that we observed.
4. **Estimator:** next year's quadrats have not been counted yet, so their mean is a random variable. Once the colleague counts them, it becomes an estimate. This is the item that leads to the next slide: the value of an estimator varies from sample to sample.

Slide 17 — The sampling distribution

The **sampling distribution** of an estimator is the distribution of its estimates across repeated hypothetical samples. A hypothetical sample is another sample that the data generating process could have produced. The design and the sample size are part of that process, so every hypothetical sample has the same design and sample size as the data, and each gives a new estimate. Because we treat the data as if they were a random sample from the model (slide 5), we obtain hypothetical samples by sampling from the model. The sampling distribution tells us how much an estimate could have changed if the data generating process had produced another sample, so we use it to quantify the uncertainty of an estimate (slides 22 and 24). It is an imaginary construct: in a real study we observe one realisation of the data generating process, obtain one estimate, and never observe the sampling distribution itself.

There are two distributions and two kinds of repetition. Within one sample, the quadrats are the replicates, and the Poisson distribution describes the variation of the counts across them. Between samples, the sampling distribution of $\hat{\lambda}$ describes the variation of the mean of the n quadrats of a sample across hypothetical samples. Adding quadrats to a sample increases n; it does not produce another sample. In the same way, if the quadrats had been counted in two field campaigns and we analysed all the counts together, they would form one sample: a single realisation of a data generating process that includes both campaigns.

For a finite sample, the shape of the sampling distribution is in general unknown, except in special cases where it can be derived mathematically. The general way to study it is a simulation, where we choose the parameter of the model and draw as many hypothetical samples as we want.

Optional remark (the chapter's footnote)

Many statistics textbooks call each hypothetical sample an "experiment" and speak of repeating the experiment. The course avoids that term: much ecological data comes from surveys and monitoring campaigns rather than experiments, and an experiment is usually understood as a unit fixed in time and space, whereas what counts as one sample depends only on the data that we analyse (it may combine several field experiments or use part of one). Students who read other textbooks will meet the term.

Figure walkthrough

The schematic has no numbers. Point to the model on the left, then to one row of quadrats (one sample, with the variation among quadrats), then to the estimate of each sample and to the histogram that gathers them (the variation among samples).

Slide 18 — Hypothetical samples in R

We assume a Poisson model with $\lambda = 2.14$ (the estimate from the fir seedlings) and samples of 50 quadrats, as in the data, and draw 2,000 hypothetical samples. For each one we calculate the estimate of λ. The seed is set once, before the whole batch of repetitions.

We repeat the sampling with `replicate()`, which evaluates an expression (`expr`) a number of times (`n`) and collects the results in a vector. Here the expression simulates the counts of one sample with `rpois()` and returns their mean, so the vector `lambda_estimates` holds one estimate per hypothetical sample. The last two lines summarise it with its mean and its standard deviation; the next slide shows the result.

Slide 19 — A simulated sampling distribution

The mean of the 2,000 estimates is close to the parameter that we gave to the simulation, and their standard deviation, about 0.205 seedlings per quadrat, measures how much the estimate changes from one sample to the next. We use both values on the next slides.

Figure walkthrough — included in the three minutes

The figure is a copy of the chapter figure. The grey histogram is the distribution of the 2,000 estimates, which approximates the sampling distribution of $\hat{\lambda}$, and the vermillion line is the parameter used in the simulation. The horizontal axis is a mean count per quadrat, which is why its values are not restricted to integers. The histogram is a simulation: with real data we would have only one of these estimates.

Slide 20 — Bias and variance

We describe a sampling distribution with its mean and its variance, which tell us whether the estimator is centred on the parameter and how much the estimates vary around that centre. The **bias** of an estimator is the difference between the mean of its sampling distribution and the parameter that it estimates; θ stands for a parameter in general and $\hat{\theta}$ for its estimator. An estimator is **unbiased** when its bias is zero.

For i.i.d. Poisson counts, $E\left[\hat{\lambda}\right] = \lambda$, so the mean of the sample is an unbiased estimator of λ (the mean of the sample is unbiased for the mean of the model for any distribution). However, unbiasedness is the exception and not the rule. Most estimators are biased in finite samples, such as the method-of-moments estimator of a variance (slide 28) or a non-linear function of an unbiased estimator, such as $e^{-\hat{\lambda}}$ as an estimator of the probability $e^{-\lambda}$ of an empty quadrat. In general, we can only expect the bias of an estimator to become small as the sample grows (slide 29). Bias is a property of the whole sampling distribution: individual estimates of an unbiased estimator still fall above or below the parameter, as the histogram of slide 19 shows.

The **variance of an estimator** is the variance of its sampling distribution. For the mean of n i.i.d. Poisson counts it is λ/n: it decreases as the sample grows. The variance of the counts in individual quadrats remains λ; what decreases is the variation among the means of hypothetical samples. The simulated values on the slide are close to these results, and they do not match exactly because 2,000 hypothetical samples are still a finite number of repetitions.

The formula λ/n is a special case of a general result: the mean of n i.i.d. observations has variance σ^2/n, where σ^2 is the variance of the model (which equals λ for a Poisson distribution). Exact formulae like this exist for only a few estimators, such as the mean. For the others, we rely on large-sample approximations (slides 29 and 30) or on simulation.

Short activity — included in the six minutes

Ask the questions on the slide (the chapter's "Practice: what varies?"). Expected answers: the samples, and therefore the estimates, vary; the parameter λ of the model is the same in both studies. Bias is the difference between the mean of the sampling distribution and the parameter, and two estimates are not enough to describe a sampling distribution (also, in a real study we would not know λ).

Slide 21 — Mean squared error

The **mean squared error** (MSE) combines bias and variance into one measure of the quality of an estimator. It is the expected squared distance between the estimator and the parameter, and it equals the variance plus the squared bias. For an unbiased estimator the MSE equals its variance, so for the mean of Poisson counts it is λ/n; in the simulation it is about 0.0418.

The MSE is useful to compare estimators of the same parameter. A biased estimator can have a smaller MSE than an unbiased one if its variance is sufficiently smaller. We see an example with two estimators of a variance on slide 28, and students compare them by simulation in Exercise 3 of the practical.

Optional remark (the chapter's footnote)

Classical statistics put a strong emphasis on unbiased estimators. Many fields of research now favour biased estimators when this substantially reduces the mean squared error; the keyword for further reading is the bias–variance trade-off.

Slide 22 — Standard errors

The **standard error** (SE) of an estimator is the standard deviation of its sampling distribution. We prefer it to the variance when reporting uncertainty because it has the units of the estimate. For the mean of Poisson counts it is $\sqrt{\lambda/n}$; in the simulation that is $\sqrt{2.14/50} \approx 0.207$, close to the standard deviation of the 2,000 estimates (0.205).

With real data we do not know λ, so we cannot calculate the standard error. Instead we replace the parameter with its estimate. This is a **plug-in estimate**: the standard error that we report is itself an estimate, and a rough one in small samples. For the 50 quadrats it is about 0.21 seedlings per quadrat. That is, across hypothetical samples of 50 quadrats each, we would expect the estimates of λ to vary with a standard deviation of about 0.21 seedlings per quadrat. It is the same value as in the simulation because we gave the simulation the estimate as its parameter.

This calculation relies on the Poisson model. If the counts are overdispersed with respect to the Poisson distribution (Chapter 2), the formula underestimates the variation across hypothetical samples; Exercise 1 of the practical shows an example with the barn owls.

Slide 23 — Discussion: standard deviation or standard error?

These two terms are often confused. This discussion separates them with the numbers of the fir seedlings.

Discussion and expected answers — included in the four minutes

Give students one minute, then take one answer per item.

1. **Standard deviation:** the variation among the observations, the counts of individual quadrats (`sd()` of the 50 counts).
2. **Standard error:** the variation of the estimate across hypothetical samples.
3. **Standard error:** the histogram of slide 19 is a simulated sampling distribution, so its spread approximates the standard error.
4. **Standard deviation:** the variation of the count of one quadrat under the model, $\sqrt{\lambda}$.

Then say the chapter's warning once: a larger sample reduces the standard error, but it does not reduce the standard deviation of the observations.

Break (10 minutes).

Slide 24 — Confidence intervals

A **confidence interval** is a range of values for a parameter, calculated from a sample with a procedure whose behaviour across repeated hypothetical samples we can describe, exactly in special cases and approximately in general. The **confidence level** (for example 95%) is the fraction of hypothetical samples for which the interval calculated with that procedure would contain the parameter. We use confidence intervals to report the uncertainty of parameter estimates in the rest of the course.

A simple procedure is available when the sampling distribution of the estimator is approximately Normal; slide 30 discusses when this is the case. An approximate 95% confidence interval is then the estimate plus and minus $z_{0.975}$ times the estimated standard error, where $z_{0.975} \approx 1.96$ is the 0.975 quantile of the standard Normal distribution (mean 0 and standard deviation 1). This quantile leaves a probability of 0.025 in each tail, hence 95% in between. This type of interval is known as a **Wald interval**.

Optional remark (the chapter's footnote)

When the observations are i.i.d. from a Normal distribution, there is an interval for the mean with exactly 95% coverage, which replaces $z_{0.975}$ by a quantile of Student's t distribution; it is a special case that does not extend to other models.

Slide 25 — Wald interval in R

The first three lines repeat the plug-in standard error of slide 22. `qnorm()` gives the 0.975 quantile of the standard Normal distribution, and the last line calculates both endpoints at once: `c(-1, 1)` subtracts and adds the same amount. The endpoints are in the comment.

It is an interval for the parameter λ, the mean count, so it says nothing about the range of counts in individual quadrats: most quadrats have counts outside it.

Slide 26 — The dance of the confidence intervals

The parameter is fixed and the interval is random: its endpoints change from one hypothetical sample to the next. If we had many hypothetical samples and calculated an interval from each with the same procedure, about 95% of those intervals would contain the parameter. Once an interval has been calculated, it either contains the parameter or it does not, so we do not say that the parameter has a probability of 95% of being inside it.

Figure walkthrough — included in the five minutes

If the connection allows, open Geoff Cumming's animation of the "dance of the confidence intervals" at https://logarithmic.net/2017/dance/ for about one minute: estimates and their intervals bounce and shift from one sample to the next. Otherwise use the static version on the slide.

The figure is a copy of the chapter figure, for the Poisson model of the seedling counts. At the top, the simulated sampling distribution of slide 19; below it, the estimates (points) and 95% Wald intervals (lines) of 20 hypothetical samples, on the same horizontal axis. The dashed line is the parameter. Nineteen intervals cross it; the sixth sample, in vermillion, has a low estimate, and its interval misses the parameter. With 20 samples, one miss is what a 95% procedure leads us to expect on average, but another set of 20 could have none or several.

Slide 27 — Coverage checked by simulation

The fraction of hypothetical samples in which the interval actually contains the parameter is called the **coverage** of the procedure. The Wald interval relies on an approximation (and on a plug-in standard error), so its coverage may differ from the confidence level that we intended. We can check this with a simulation, where we know λ: for each simulated sample we calculate the interval and record whether it contains the parameter. The code is on the chapter page.

With a mean of 2.14 and 50 quadrats, the simulated coverage is very close to the intended 0.95. This result is specific to these values. When the expected total count $n\lambda$ is small, the sampling distribution of $\hat{\lambda}$ is far from Normal and the coverage of the Wald interval can be noticeably lower than 95%. The lower endpoint can even be negative, which is not a possible value for the mean of a Poisson distribution.

Short activity — included in the four minutes

Ask the questions on the slide (the chapter's "Practice: interpret an interval"). Expected answers: the 95% refers to the coverage of the procedure across hypothetical samples: about 95% of the intervals calculated in this way would contain the parameter. For a Wald interval based on a Poisson model, we would check whether the counts can be treated as i.i.d. Poisson (for example, whether they are overdispersed) and whether the sample is large enough for the Normal approximation.

Slide 28 — Estimation of two parameters

The concepts above apply in the same way to continuous measurements and to models with more than one parameter. We return to the flipper lengths and assume that they are i.i.d. Normal, where μ is the mean of the model (in mm) and σ^2 its variance (in mm²). Remember that the argument `sd` of `rnorm()` and related functions takes the standard deviation σ.

The model has two parameters, so the method of moments needs two equations. The mean gives $\hat{\mu} = \overline{X}$ as before. The variance of the model is σ^2, and its counterpart in the empirical distribution is the mean of the squared deviations of the flipper lengths from the mean of the sample: the equation on the slide, with divisor n. That is, the method-of-moments estimators of μ and σ^2 are the mean and the variance of the empirical distribution (slide 10).

For the 123 penguins the two estimates of the variance differ, and these estimates are behind the rounded values of 217 mm and 6.5 mm that we have used for the Normal model since Chapter 2. We now explain the difference that we found on slide 10. The method-of-moments estimator divides the sum of squared deviations by n, whereas `var()` divides it by $n - 1$. For i.i.d. observations, the method-of-moments estimator of the variance is biased: on average it underestimates σ^2, by a fraction $1/n$ of it. This happens because the deviations are calculated from $\overline{X}$, which is estimated from the same sample and is therefore closer to the observations than μ. Dividing by $n - 1$ removes the bias, and that is the estimator that most people call the sample variance. This unbiasedness does not carry over to transformations: the square root of the result of `var()`, which is what `sd()` calculates, is a biased estimator of σ.

The bias of the method-of-moments estimator decreases as the sample grows. We do not know σ^2, but if we plug in the estimate of 42.1 mm², the bias would be about -0.34 mm² for a sample of 123 penguins and about -1.68 mm² for a sample of 25 penguins. Being unbiased does not imply having the smallest MSE: in Exercise 3 of the practical, students compare the MSE of the two estimators by simulation.

Optional remark (the chapter's box "Statistical detail: the bias of the variance estimator")

The $n - 1$ divisor is known as Bessel's correction. It was originally proved by Gauss in 1823 and popularised by Bessel in a paper from 1838. The current explanation, that we remove one degree of freedom because the estimate of the variance relies on the estimate of the mean and not on the mean of the model, is due to Fisher, about a hundred years later.

Slide 29 — Consistency

On slide 10 we saw that the mean, variance and quantiles of the empirical distribution converge to those of the model as the sample grows. Method-of-moments estimators are calculated from the moments of the empirical distribution, so they inherit this behaviour. An estimator is **consistent** if its estimates are increasingly unlikely to miss the parameter by any given margin as the sample grows. This is the same statement that we made for relative frequencies on slide 8, now applied to an estimator, and it should be read in the same way: it is about probabilities and not about every sample. The mean of i.i.d. Poisson counts is a consistent estimator of λ. The method-of-moments

estimator of the variance of the Normal model is also consistent, although it is biased for any finite sample. The formal statement is in the chapter's box "Statistical detail: consistency".

The standard error tells us how fast an estimator approaches the parameter. For the mean of Poisson counts it is $\sqrt{\lambda/n}$, so we need a sample four times larger to halve the standard error.

Short activity — included in the three minutes

Ask the question on the slide (the chapter's "Practice: two studies of different size"). Expected answer: no. Consistency states that the probability of an error larger than any given margin decreases towards zero as the sample grows. It does not state that every larger sample is closer to the parameter than every smaller one, so the smaller study can produce the closer estimate by chance.

Slide 30 — Asymptotic normality

Large samples have a second consequence. As the sample grows, the sampling distribution of most estimators approaches a Normal distribution centred on the parameter, whatever the distribution of the individual observations. An estimator with this behaviour is said to be **asymptotically normal**. This is a general behaviour of estimators and not a special feature of the mean of a sample, although the mean is the case with its own name: for the mean of n i.i.d. observations from a distribution with finite variance, the result is the **central limit theorem**. For the mean of Poisson counts, the approximating Normal distribution has the parameter λ as its mean and the standard error $\sqrt{\lambda/n}$ as its standard deviation, which is the equation on the slide. The formal statement is in the chapter's box "Statistical detail: the central limit theorem".

Other method-of-moments estimators behave in the same way. The moments of the empirical distribution are themselves means of i.i.d. quantities, so the theorem applies to each of them. The estimate of a parameter is calculated from these moments with a smooth function, and in a large sample the moments vary so little that the function is nearly a straight line over that range, which preserves the Normal shape. So the method-of-moments estimator of the variance of the Normal model is asymptotically normal, and so are the estimators of the Negative Binomial model in the practical. Chapter 4 gives the same result for maximum likelihood estimators, with a way to calculate their standard errors.

Asymptotic normality is the justification for the Wald interval of slide 24, and because it is a general behaviour the same interval can be used with estimators other than the mean. How large n needs to be depends on the model and on the estimator. For Normal observations the sampling distribution of the mean is exactly Normal at any sample size, whereas that of the method-of-moments estimator of the variance has a skewness of approximately 1.4 for 5 observations and is still slightly skewed for 123 (approximately 0.26).

Figure walkthrough — included in the five minutes

The figure is a copy of the chapter figure: 2,000 simulated means for samples of 5 and of 100 quadrats, each with its Normal approximation. For 5 quadrats the sampling distribution is discrete (the mean of five counts can only change in steps of 0.2, so each bar is one possible value) and slightly asymmetric, with a longer tail to the right. For 100 quadrats the Normal approximation is

close. The counts in individual quadrats still follow a Poisson distribution in both cases; it is the distribution of their mean that approaches a Normal distribution.

Two cautions, said once

The chapter's warning "Finite samples and special cases": in a finite sample an estimator is in general biased and the shape of its sampling distribution is unknown. Exact results, such as the t interval for the mean of Normal observations, exist for only a few estimators and models. For the models of this course we can make confident statements about bias, variance, standard errors and confidence intervals only for large samples; for small samples they are approximations, and simulation is the tool to check how good they are.

Consistency, coverage and asymptotic normality are all derived under the assumed model, in our case i.i.d. Poisson counts. In a real forest the expected count may change with habitat, seedlings may be clustered, or some seedlings may be missed. A larger sample does not correct for this. Instead, we need a model that describes those sources of variation, such as the distributions for overdispersed counts of Chapter 2 or the models for grouped observations of Chapter 7.

Slide 31 — Discussion: empirical or sampling distribution?

This discussion separates the two distributions of the lecture.

Discussion and expected answers — included in the three minutes

Give students one minute, then take one answer per item.

1. **Empirical distribution:** a histogram of individual observations.
2. **Sampling distribution**, approximated by simulation: each value in the histogram is one estimate calculated from a whole sample of 50 quadrats.
3. **Empirical distribution:** the relative frequencies of the observed counts.
4. **Sampling distribution:** the standard error is its standard deviation.

Slide 32 — Course roadmap: from samples to response curves

In this chapter we learned how a sample relates to a model and how estimates vary across hypothetical samples. Chapter 4 replaces the method of moments by maximum likelihood, a general method of estimation that we use in the rest of the course. The estimators change, but we describe their uncertainty with the concepts of today: standard errors, confidence intervals, coverage and asymptotic normality. Chapter 5 turns to the deterministic functions that describe ecological responses, which Chapter 6 combines with the distributions. The observations that we treated as i.i.d. as a simplification because they come in groups (visits to the same nest) come back in Chapter 7.

Slide 33 — The Chapter 3 R practical

The practical has four exercises. The first, on barn owl calls, is worked on the page, so read it first: a Negative Binomial model fitted by the method of moments, a standard error and a Wald interval. It uses the second and third blocks together and shows how the practical works. The Poisson formula would give a standard error of about a third of the Negative Binomial one, which

is the warning of slide 22. The 122 visits come from only 24 nests, so treating them as i.i.d. is a simplification that we accept in this chapter; Chapter 7 covers observations nested in groups. Exercise 2 simulates samples of 10 and 1,000 leaves from a Beta model of leaf damage, with several seeds, to see how the properties of a sample behave as the sample grows, and compares one large sample with the model in a histogram and a cumulative plot. Exercise 3 uses the heights of 31 black cherry trees: two estimators of the variance, compared by simulation for bias and mean squared error. Exercise 4 simulates the body mass of small mammals under a LogNormal model and checks the coverage of Wald intervals for the mean mass at two sample sizes.

Read the Beta and LogNormal entries of the supplement first: they give the mean, the variance and the R functions that the exercises need. Try each exercise before checking the solution.

Slide 34 — Summary and closing questions

Go back to slide 4 by speaking about it again; no new slide is needed. We treat a sample as random draws from a model, and we compare the sample with the model through relative frequencies, histograms and empirical cumulative distributions. An estimator is a rule that calculates a value for a parameter from a sample, and an estimate is its result for one sample. The sampling distribution of an estimator describes how its estimates vary across hypothetical samples: its mean gives the bias, its standard deviation is the standard error, and the mean squared error combines both. A confidence interval comes from a procedure that contains the parameter in a given fraction of hypothetical samples, and we can check that coverage by simulation. Consistency and asymptotic normality describe large samples; for small samples we check by simulation. On the diagram of slide 4, the dashed arrow from the estimates to the parameters now comes with a standard error and an interval.

Return to this slide for the two closing questions.

Discussion and expected answers — allow about two minutes

Question 1: the standard deviation of the counts describes the variation among individual quadrats, which is a property of the process and does not depend on how many quadrats we count ($\sqrt{2.14} \approx 1.46$ seedlings per quadrat under the model). The standard error describes the variation of the mean across hypothetical samples; averaging more quadrats cancels more of the variation among them, so it shrinks as $\sqrt{\lambda/n}$.

Question 2: the 95% describes the procedure: across hypothetical samples, about 95% of the intervals calculated in this way contain the parameter, if the coverage of the procedure equals its confidence level. One calculated interval either contains the parameter or it does not.

Final transition

The next steps are the practical and Chapter 4, where maximum likelihood replaces the method of moments and the same concepts describe the uncertainty of the estimates.

Adjustments during delivery

The slides take 125 minutes and leave 5 minutes of slack, which comes last. If discussion overruns, use the slack first, then shorten in this order:

- Slide 3 to one sentence ("this lecture is the first step of estimation"): recovers about half a minute.
- Slide 13 merged into speech on slide 12 (one sentence on `ecdf()` and the 88% at 225 mm): about two and a half minutes.
- Slide 21 to one sentence on slide 20 (the MSE is the variance plus the squared bias): about two and a half minutes.
- Slides 7 and 18 to their definition box (the `factor()` levels; `replicate()`): about two minutes each.
- Slide 32 to one sentence (Chapter 4 replaces the method of moments by maximum likelihood): about one and a half minutes.

Keep the explanations of the two kinds of repetition (slide 17), standard deviation against standard error (slides 22 and 23), what 95% means (slides 26 and 27), and that all the properties are derived under the assumed model (slide 30).

The breaks are placed between the blocks, after slides 13 and 23 (minutes 44 and 96). If a block runs over, shorten the block rather than the following break, and catch up with the adjustments above.

If time is left: code to run

All of these chunks are visible on the chapter page or the practical page, so students can follow along and nothing needs to be typed. Start with the chunks that load `fir`, RTMB and RTMBdist, and `penguins`, and keep one R session open across the breaks, because later chunks use their objects.

- **After slide 13** (*Samples and distributions*, *Samples of continuous measurements*): the `table(factor())` chunk (slide 7), the convergence loop with seed 123 (slide 8), the `quantile()` chunk (slide 10), and the bin, `density()` and `ecdf()` chunks (slides 11 to 13). "Practice: fruit-set frequency" is done by hand ($15/24 = 0.625$). "Practice: orchid flowering across plots" (seed 246) goes last, because it redefines `count_values`.
- **After slide 23** (*Parameters, estimators and estimates*, *Sampling distributions*, *Standard errors*): the `replicate()` chunk with seed 401 (slide 18), the bias, variance and mean squared error chunks (slides 20 and 21) and the plug-in standard error (slide 22). Two practices are done by hand: "Practice: estimate from quadrats" (3 seedlings per quadrat) and "Practice: change the sample size" (0.207 becomes 0.103 seedlings per quadrat with 200 quadrats).
- **After slide 31** (*Confidence intervals*, *Estimation of two parameters*, then the practical page): the Wald interval (slide 25, after the plug-in standard error), the coverage chunk with seed 402 (slide 27), the two estimates of the flipper variance (slide 28) and Exercise 1 on barn owl calls.